PREDICTIVE MODELING OF MATERIAL TAKE-OFF OVERRUN RISKS IN BROWNFIELD EPC PROJECTS

A Dual-Task Machine Learning Approach to Quantifying MTO Growth for Proactive Cost Control

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The Brownfield Engineering Challenge

The Reality of Revamps

Modifying active, aging plants presents massive logistical hurdles compared to greenfield projects.

The Constraint Triangle

- Incomplete legacy blueprints
- Tight physical and spatial constraints
- High tie-in density to operating headers

The Business Impact

Initial Material Take-Off (MTO) estimates drawn from early Front-End Engineering Design (FEED) rarely match physical reality, leading to severe volume growth as designs mature.

The Flaw in Flat Contingencies

Project controls teams typically apply a static, across-the-board **10% to 15%** material buffer to hedge against volume growth.

This treats benign, straight-run utility pipes identically to high-pressure, severe-temperature, or toxic exotic alloy circuits.

The Consequences of Getting it Wrong:

Over-estimating: Traps millions in idle working capital, bloating EPC bid prices and making proposals uncompetitive.

Under-estimating: Causes catastrophic field shortages, triggering high-premium air freight and expensive craft standby delays.

Project Objectives: Transforming Data into Risk Intelligence

Replacing subjective markups with an empirical, data-driven machine learning pipeline.

Flag High-Risk Overruns (Early Warning): Identify specific process subsystems that have a high probability of exceeding a 15% material growth before the design is frozen.

Quantify Exact Material Growth (Smart Shopping List): Predict the exact percentage volume overrun based on process severity and spatial constraints.

Optimize Contingency Allocation: Empower project managers to justify dynamic, risk-adjusted material allowances to protect both the schedule and the bottom line.



A Dual-Task Machine Learning Framework

Trained on 8,500 domain-synthesized brownfield subsystems mimicking real-world constraints.

Task 1: Classification (Risk Flagging)

Model: Champion LightGBM Classifier

Result: Achieves a 97.41% PR-AUC score.

Value: Highly sensitive to predicting "danger zones" without triggering excessive false alarms.



Task 2: Regression (Growth Quantification)

- **Model:** Champion LightGBM Regressor
- **Result:** Restricts residual variance to ~3.5% (RMSE).
- **Value:** Provides a mathematically proven uncertainty band, drastically outperforming standard linear estimating techniques.



Business KPI 1: Proactive Cost & Schedule Control



Sharper Bid Competitiveness

- Unnecessary contingency padding on low-complexity utility circuits (e.g., carbon steel, 100% laser-scanned) is pruned, allowing linear, more competitive LSTK bids.

Targeted Protection for Critical Paths

- Capital is disproportionately redirected to high-severity, exotic alloy lines (Duplex, Inconel).

Schedule Protection

- Prevents field shortages that trigger vendor expedited shop rates and craft standby delays.

Business KPI 2: Optimizing Working Capital

Capital Allocation on a $20M Bulk Piping Package

Conventional 15% Blanket Contingency:	ML-Adjusted Allowance (±3.5% Uncertainty Band):	Net Working Capital Released:
$3M allocated as a static hedge.	$0.7M to $1.4M allocated dynamically per line item.	Frees up $1.6M to $2.3M in trapped working capital per $20M package.

The “Glass Box” Approach for Engineers

Model logic is fully transparent to technical audit boards using Local SHAP Explanations.

- The model relies on physical engineering laws, not black-box chance.

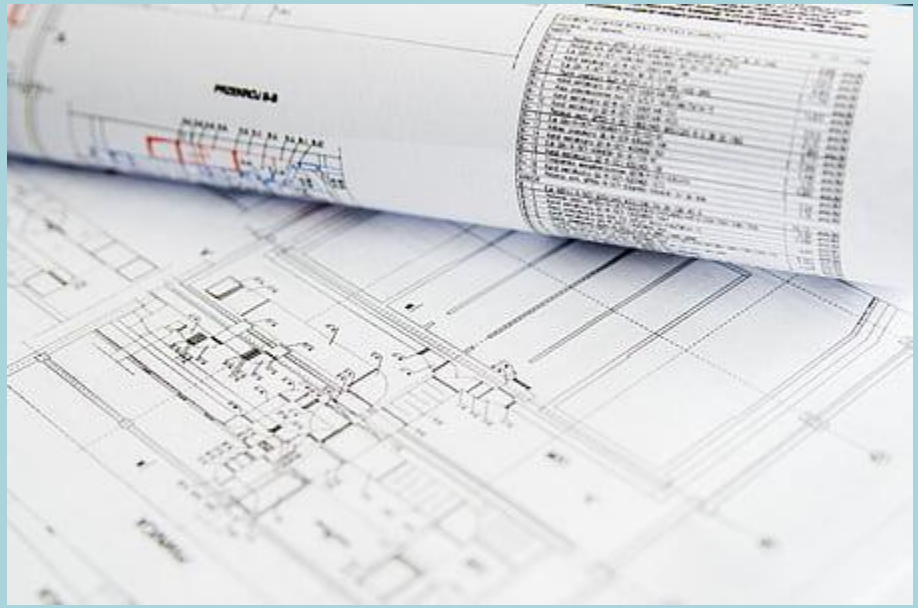
- **Top Drivers of Overruns:**
  1. High Brownfield Complexity Penalty (dense tie-ins)
  2. Lack of 3D Laser Scan Coverage
  3. Low FEED Gate Maturity (30% vs. 90%)

The Verdict: The model mathematically confirms that spatial uncertainty and the early design stage are the largest contributors to bulk material growth.



Local Explainability (Line PL-00005 Audit)

Tracing every dollar of contingency back to a physical constraint.



Historical Baseline Material Growth: 19.37%

Risk Escalators Identified by ML:

- 30% FEED maturity (+5.16% penalty)
- Toxic fluid service (+3.23% penalty)
- Extreme design pressure of 240.9 barg (+1.86% penalty)

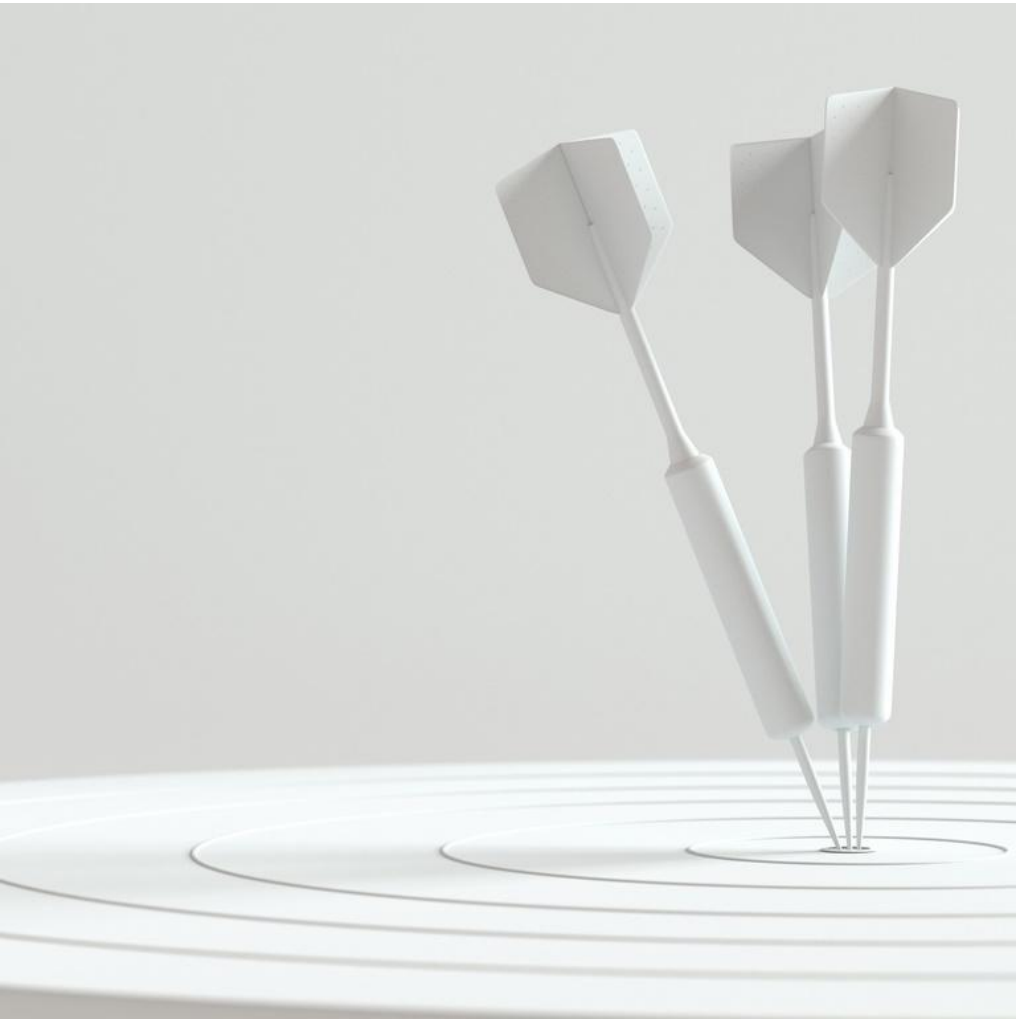
Risk Mitigators Identified by ML:

- 97.3% 3D laser scan coverage (-1.67% credit)

Final ML Verdict: Targeted **27.73%** buffer, precisely matching the actual historical closeout need of 27.54%. A traditional 15% blanket markup would have left this critical line severely under-budgeted.

Algorithmic Fairness & Unbiased Estimation

- Evaluates engineering physics (pressures, tie-ins), not human or commercial metadata.
- Validated using Equalized Error Distribution (Group Residual Parity) across subgroups.
- Zero-centered error distributions guarantee statistical parity.
- The algorithm does not penalize outsourced contractors or regional suppliers, ensuring completely equitable bid allocations.



Thank you

Special thanks to the AIM faculty, program leader, and my cohort for their guidance throughout this program journey.

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